UIDAI Aadhaar Enrollment Analysis – 2025

- **Project Title:** UIDAI Aadhaar Enrollment Analysis – 2025
- **Focus:** Trends across regions, districts, and age groups in India
- **Participant:** Ritika Patil (Solo Entry)
- **Hackathon:** UIDAI Data Hackathon 2026
- **Timeframe Analyzed:** March – December 2025

Why Analyze Aadhaar Enrollment Trends?

Aadhaar enrollment is a cornerstone of India’s digital governance system. Understanding enrollment trends helps identify regions, districts, and age groups that require targeted attention. This analysis focuses on temporal patterns, regional hotspots, and age-wise distribution to provide actionable insights that can guide UIDAI’s outreach programs and policy planning.

Dataset Overview

- **Source:** UIDAI Aadhaar Enrollment Dataset
- **Number of Records:** ~1,006,029
- **Columns:** State, District, Age groups (0–5, 5–17, 18+), Date
- **Files:** 3 CSV files merged
- **Time Period:** March – December 2025
- **Format:** Raw CSV files (aggregated during analysis)

```
df1 = pd.read_csv("/content/api_data_aadhar_enrolment_0_500000.csv")
df2 = pd.read_csv("/content/api_data_aadhar_enrolment_1000000_1006029.csv")
df3 = pd.read_csv("/content/api_data_aadhar_enrolment_500000_1000000.csv")
```

```
df = pd.concat([df1, df2, df3], ignore_index=True)
df
```

	date	state	district	pincode	age_0_5	age_5_17	age_18_greater
0	02-03-2025	Meghalaya	East Khasi Hills	793121	11	61	37
1	09-03-2025	Karnataka	Bengaluru Urban	560043	14	33	39
2	09-03-2025	Uttar Pradesh	Kanpur Nagar	208001	29	82	12
3	09-03-2025	Uttar Pradesh	Aligarh	202133	62	29	15
4	09-03-2025	Karnataka	Bengaluru Urban	560016	14	16	21
...	...	...	...	...	...	...	...
1006024	31-12-2025	Telangana	Hyderabad	500045	4	5	1
1006025	31-12-2025	Telangana	Hyderabad	500057	0	2	0
1006026	31-12-2025	Telangana	Hyderabad	500061	4	2	0
1006027	31-12-2025	Telangana	Hyderabad	500062	1	4	0
1006028	31-12-2025	Telangana	Hyderabad	500095	0	1	0

1006029 rows × 7 columns

Analysis Approach & Workflow

Step 01

Data Acquisition & Integration

- Aadhaar enrollment datasets provided through the UIDAI Data Hackathon were collected.
- Multiple CSV files were merged into a single dataset to enable unified analysis.

Step 02

Data Cleaning & Preparation

- Date fields were converted into a standard datetime format.
- Monthly values were extracted from dates.
- Total enrollment was calculated by summing age-group values (0–5, 5–17, 18+).

Step 03

Data Aggregation & Analysis

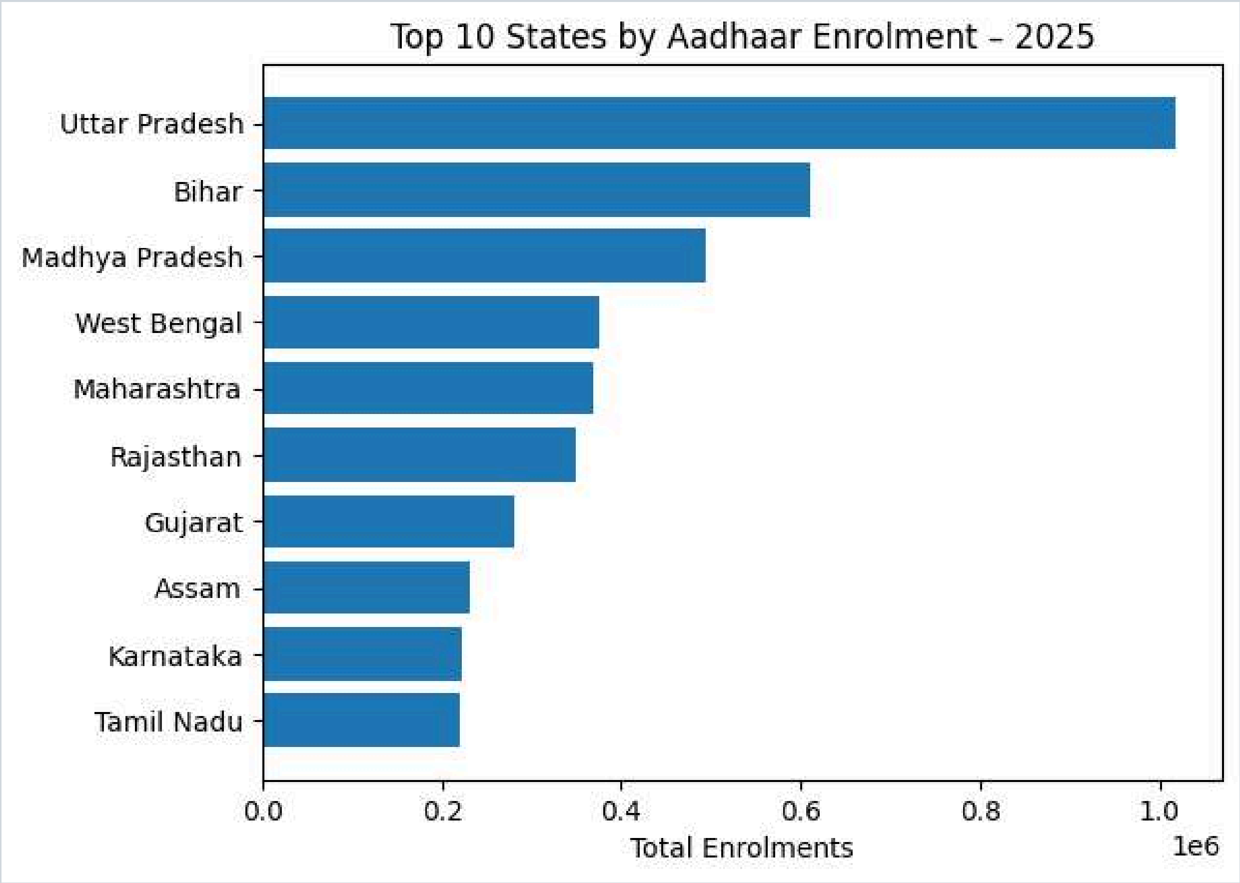
- Monthly enrollment trends were computed to identify temporal patterns.
- State-wise and district-wise aggregations were performed to detect regional hotspots.
- Age-group-wise totals were analyzed to understand demographic distribution.

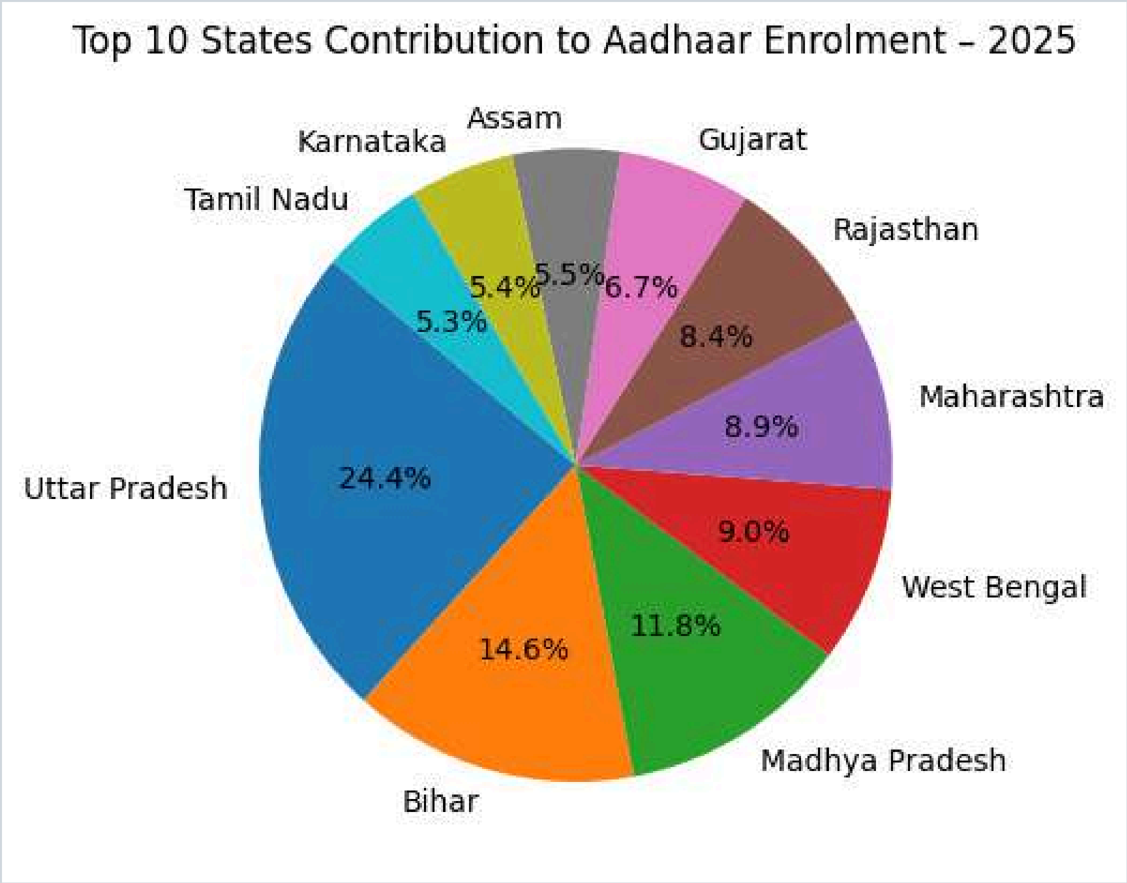
Step 04

Visualization & Insight Generation

- Line charts were used for monthly trends.
- Bar charts highlighted top states, districts, and age groups.
- Visual insights were derived to support data-driven decision-making.

Top 10 States by Aadhaar Enrollment



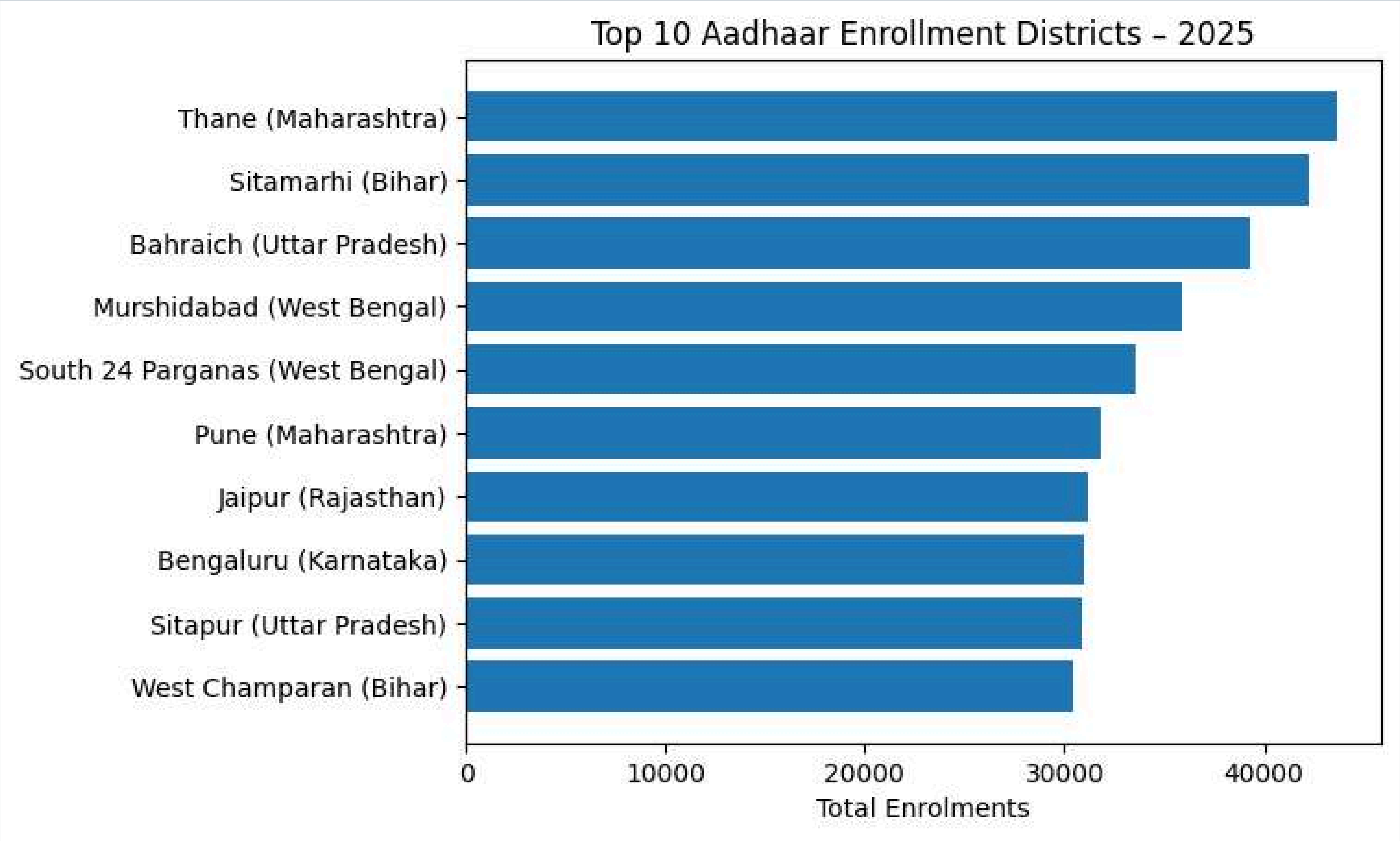


- State-wise aggregation was performed to analyze Aadhaar enrollment across different regions.
- The bar chart compares total enrollment volumes among the top contributing states.
- The pie chart represents the percentage share of each state in overall enrollments.
- A small number of states contribute a major proportion of total Aadhaar enrollments.

This analysis helps identify priority states for policy focus and outreach planning.

District-Level Aadhaar Enrollment Hotspots

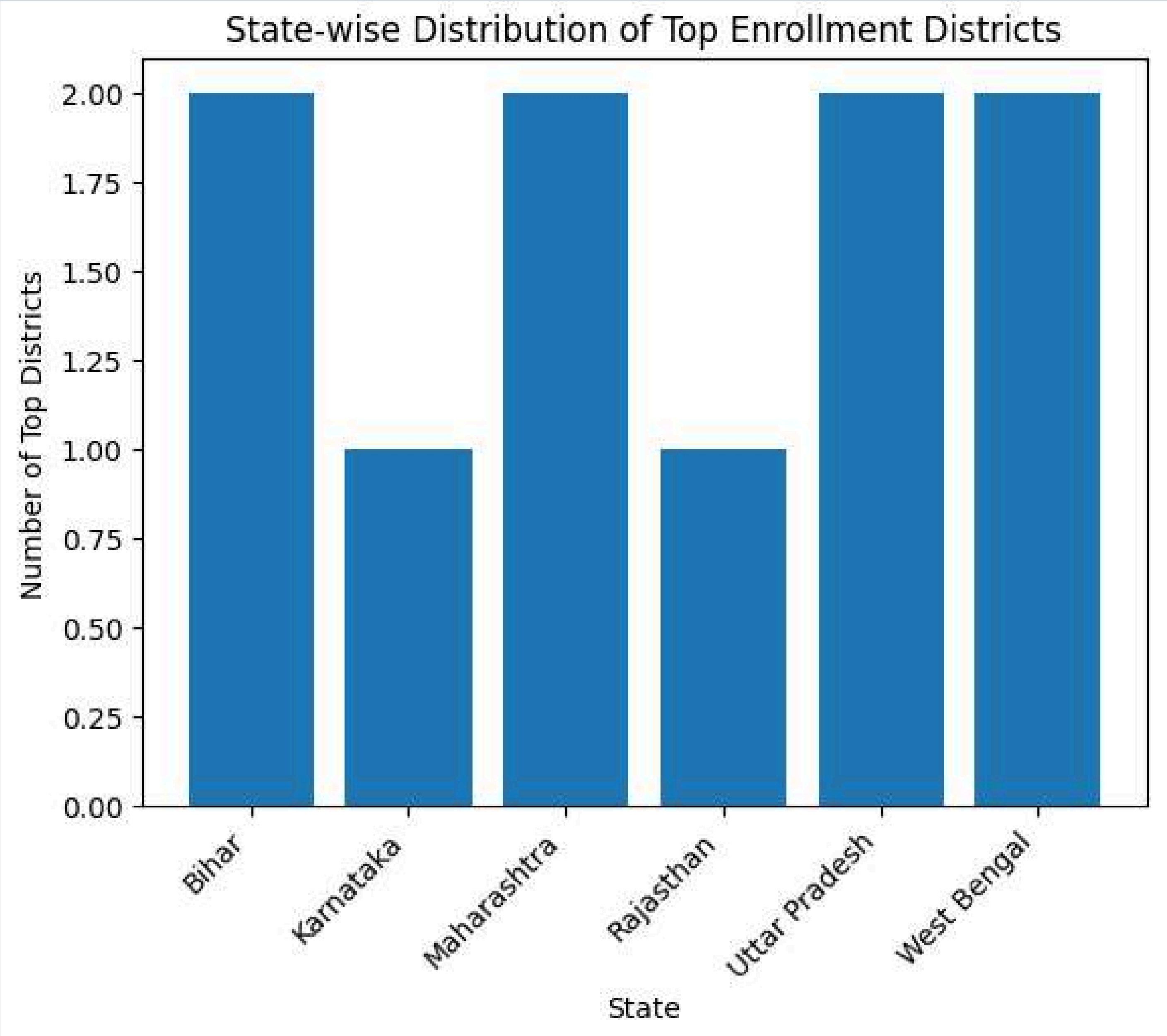
- District-wise aggregation was conducted to identify high-enrollment hotspot districts.
- The visualization highlights the top 10 districts by Aadhaar enrollment volume.
- These districts indicate areas of high registration activity or population concentration.
- Hotspot identification supports district-level monitoring and resource allocation.



- Mapping these districts enables focused and localized outreach initiatives.

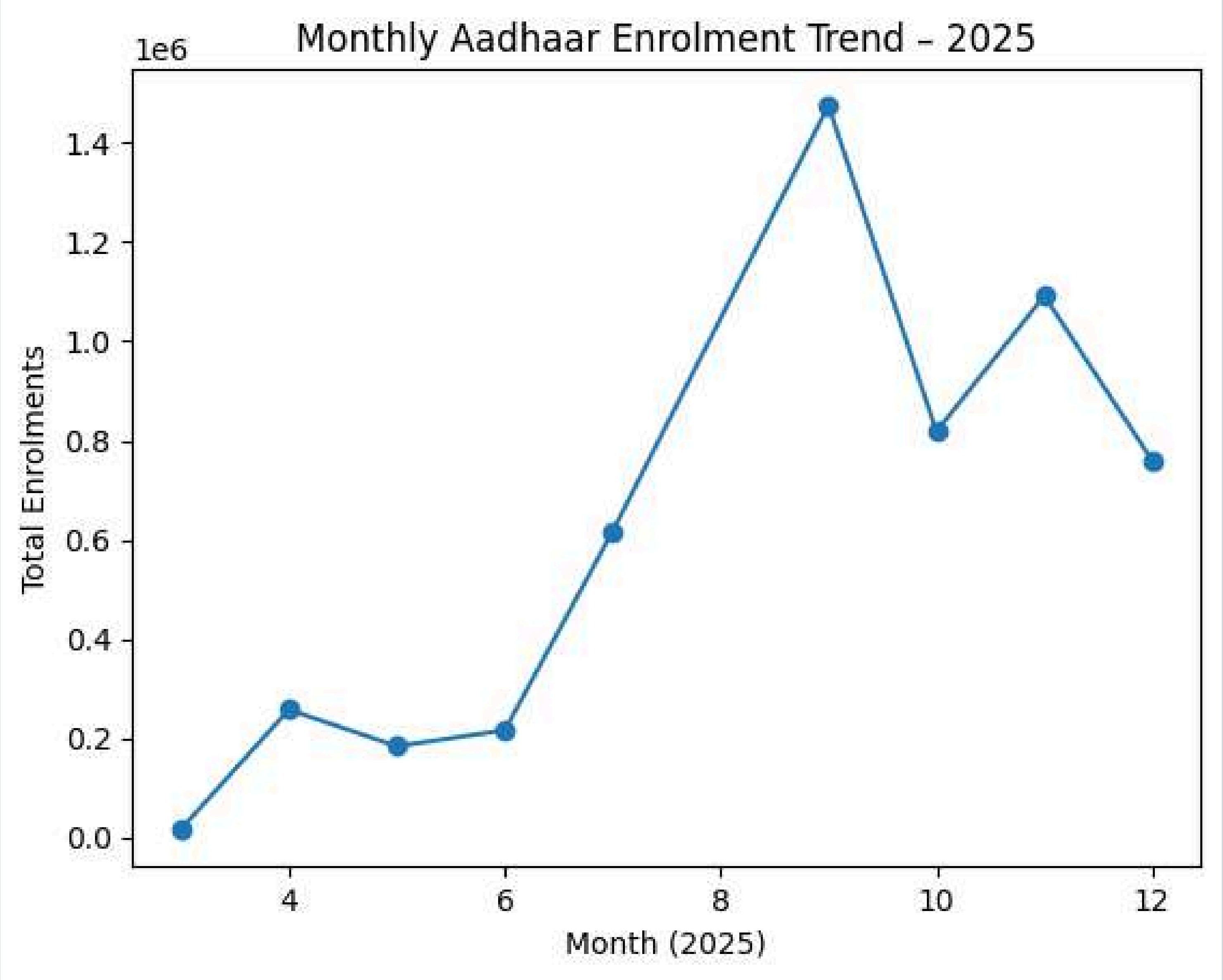
District Count per State (Hotspot Density)

- This graph represents the number of high-enrollment districts within each state.
- States with higher district counts indicate wider and more distributed enrollment activity.
- States with fewer hotspot districts show localized concentration of enrollments.
- District-level density helps identify regional demand intensity.
- This analysis is useful for state-wise administrative planning and monitoring.



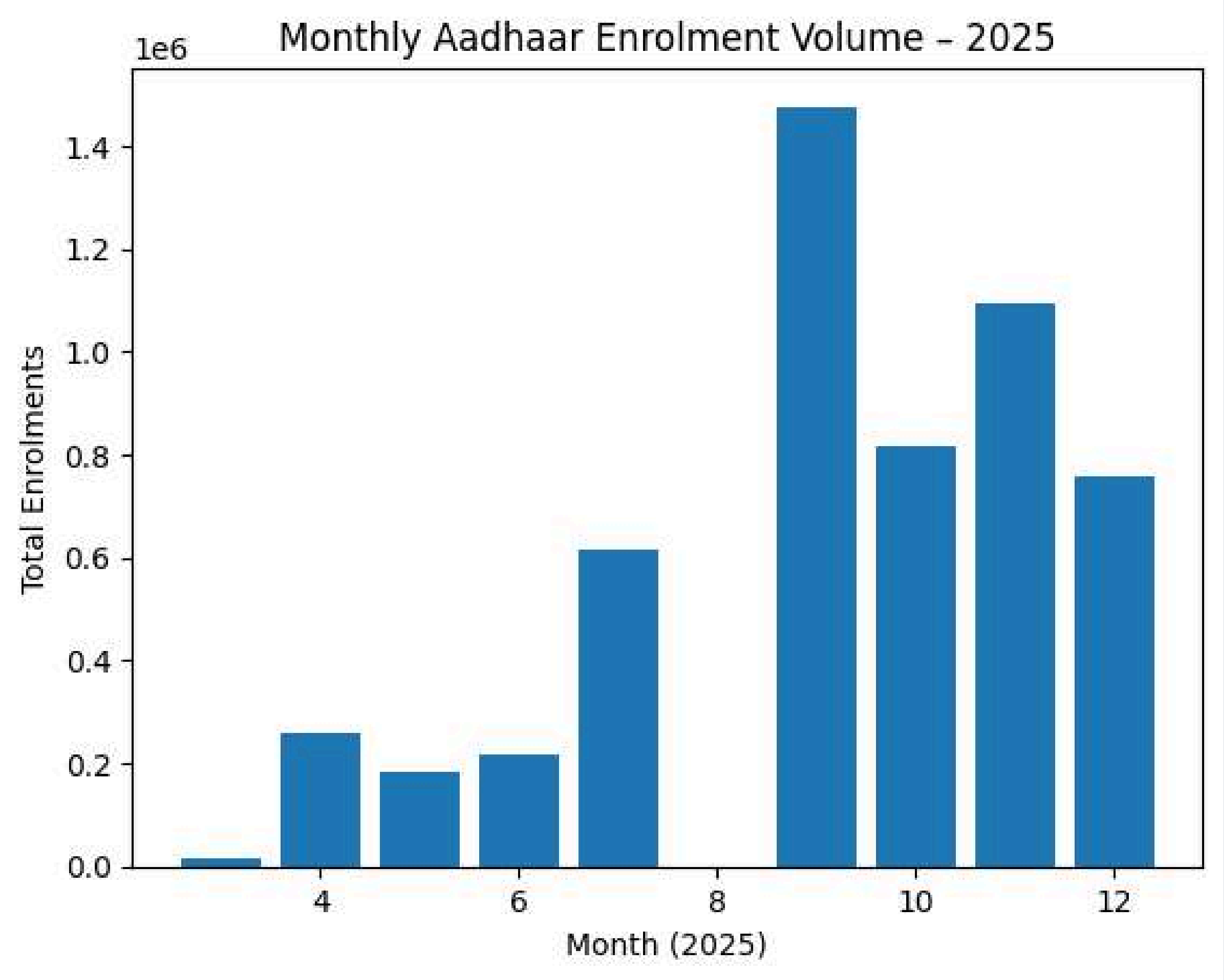
Monthly Enrollment Trends

- Month-wise aggregation was performed to study enrollment trends from March to December 2025.
- The line chart shows clear variation in enrollment volumes across months.
- Peak enrollment activity is observed during August to October, indicating periods of intensified registration drives.
- Lower enrollment in early months suggests seasonal or operational influences.
- Monthly trend insights help in planning optimal timing for outreach and enrollment campaigns.

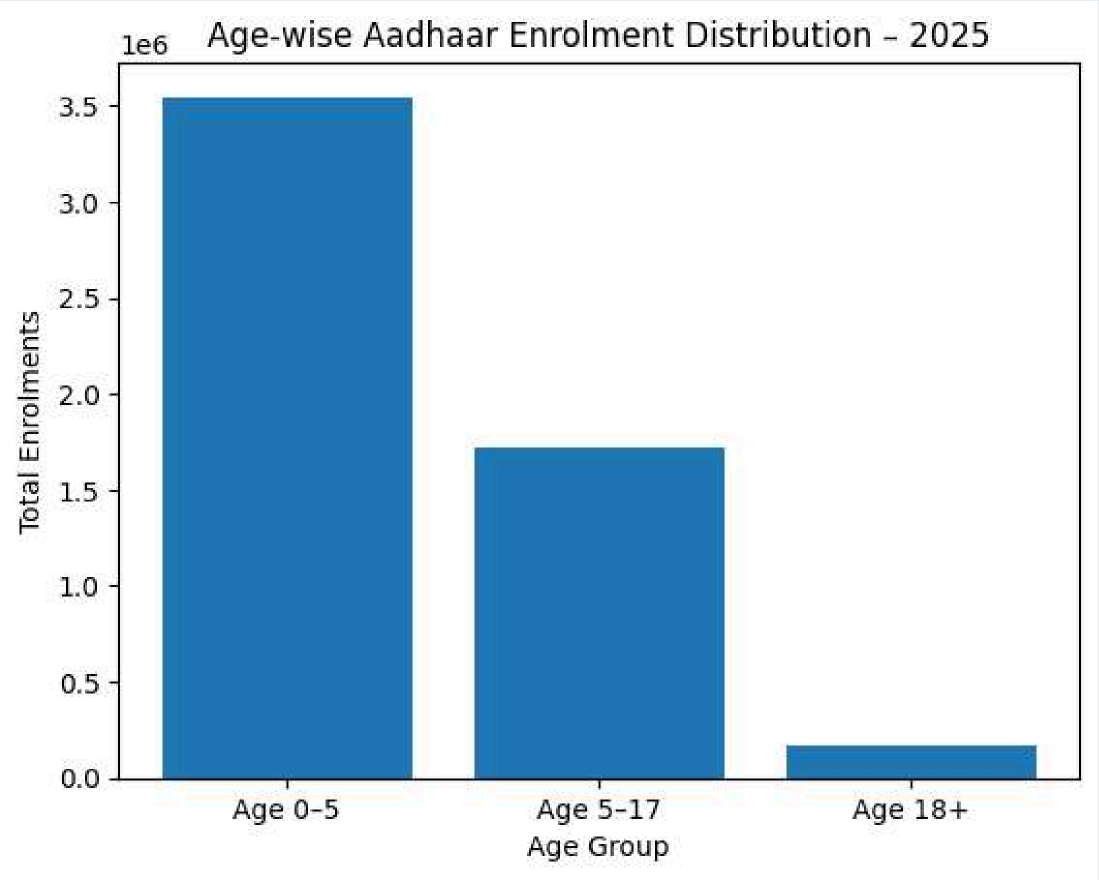


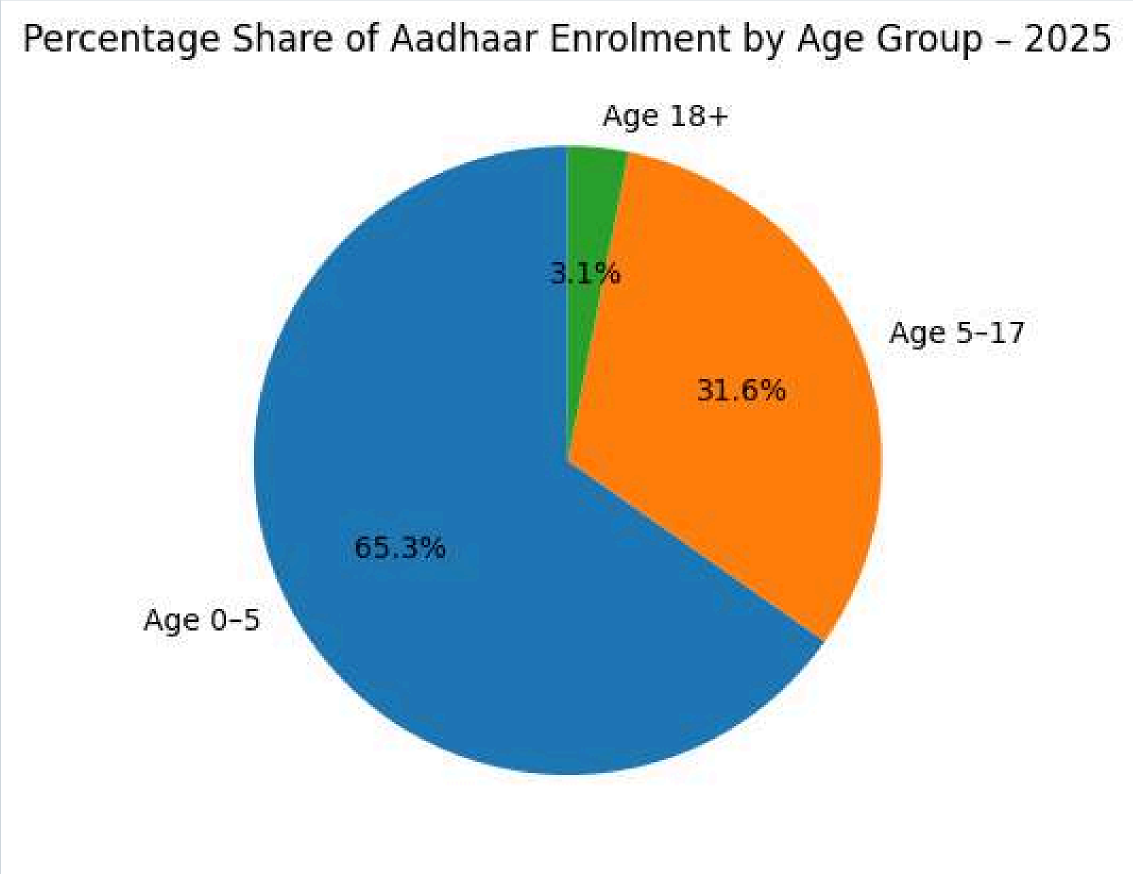
Monthly Aadhaar Enrollment Volume

- This visualization shows the absolute number of Aadhaar enrollments recorded each month in 2025.
- A sharp rise in enrollments is observed during specific months, indicating peak activity periods.
- Some months record comparatively lower volumes, suggesting seasonal or operational variation.
- The bar chart complements trend analysis by showing actual enrollment magnitude.
- Such volume-based analysis supports capacity planning and workforce allocation.



Age-wise Aadhaar Enrollment Distribution





- Enrollment data was categorized into 0–5, 5–17, and 18+ age groups.
- The bar chart shows absolute enrollment counts for each age group.
- The pie chart highlights the proportional contribution of each age group.
- The 0–5 age group accounts for the largest share of enrollments.

Age-wise insights support targeted awareness campaigns and enrollment strategies.

Key Insights from the Analysis

- Aadhaar enrollment activity varies significantly across states, districts, months, and age groups.
- A limited number of states contribute a large share of total enrollments, showing regional concentration.
- District-level analysis reveals uneven hotspot distribution, indicating localized demand patterns.
- Monthly trends highlight distinct peak enrollment periods, reflecting operational or policy-driven factors.
- The 0–5 age group dominates enrollments, emphasizing the focus on early-age registration.

Recommendations for UIDAI

01 Strengthen enrollment infrastructure during high-demand months to avoid capacity bottlenecks.

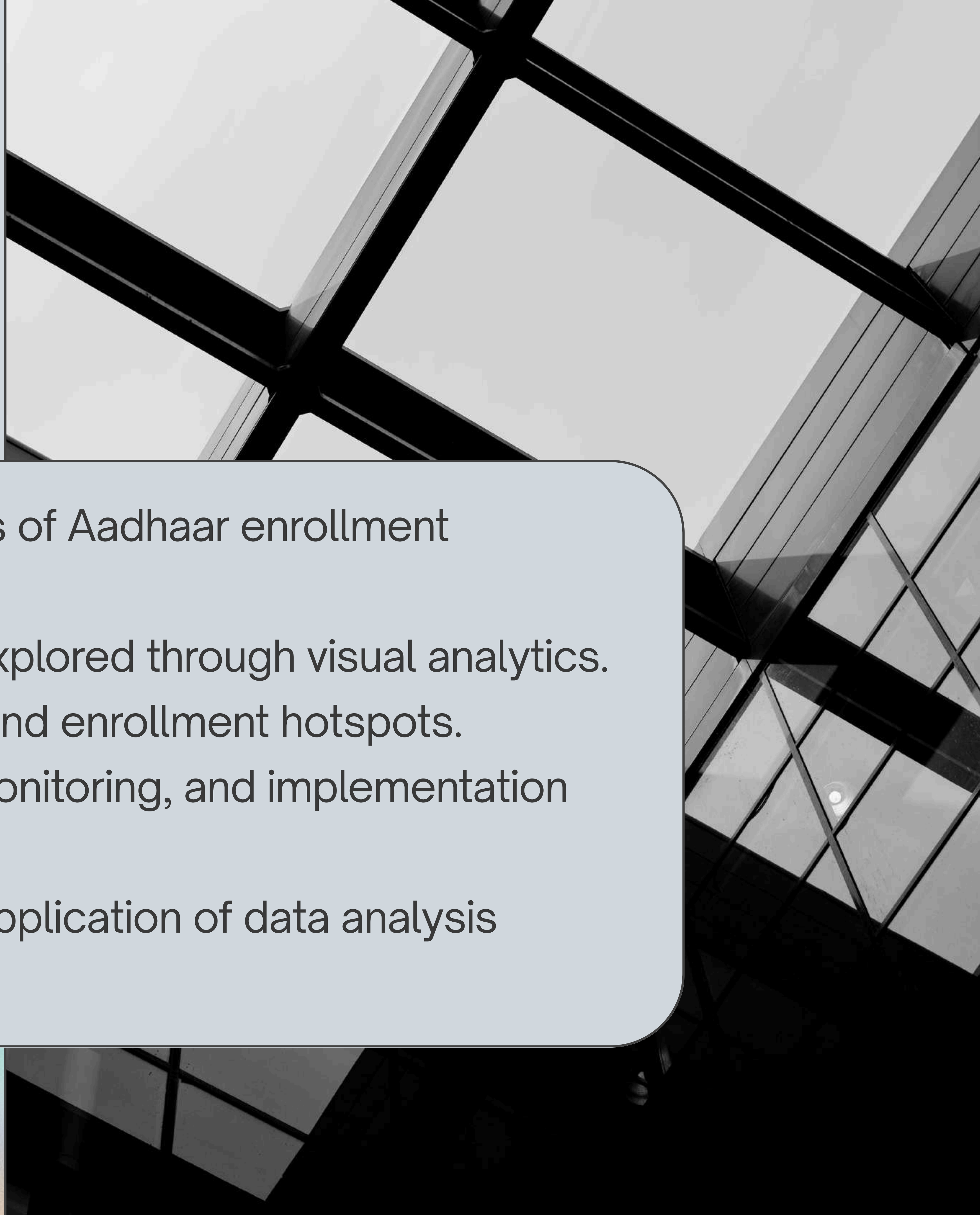
02 Increase targeted outreach in low-performing states and districts to improve coverage.

03 Continue focused initiatives for early-age (0–5) Aadhaar enrollment.

04 Use hotspot density analysis for optimized resource and workforce allocation.

05 Leverage periodic data analysis to support timely, evidence-based policy decisions.

Conclusion



- This project presents a comprehensive analysis of Aadhaar enrollment patterns using real data.
- Both temporal and regional dimensions were explored through visual analytics.
- The analysis highlights key trends, disparities, and enrollment hotspots.
- Insights derived can support better planning, monitoring, and implementation by UIDAI.
- Overall, the study demonstrates the practical application of data analysis techniques.

Future Scope & Extensions

- The project can be extended with interactive dashboards for real-time monitoring.
- Additional demographic or regional breakdowns can provide deeper insights.
- Advanced analytics may help in predicting future enrollment trends.
- This analysis supports UIDAI’s goal of efficient, inclusive, and data-driven enrollment management.

Thank You

- This project presented a comprehensive analysis of Aadhaar enrollment data using practical data analysis techniques.
- The analysis aligns with the objective of efficient, inclusive, and evidence-based enrollment management.

UIDAI Aadhaar Enrollment Analysis
Turning public data into meaningful insights